Jordan Lee

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Hillcrest, ST | US Citizen | Willing to relocate

Experience

Hardware Engineering Intern, Example Avionics, Northgate, ST May 2025–Aug 2025

- Cut noise on a 16-bit ADC front end from 12 LSB to 3 LSB by redesigning the anti-aliasing filter and grounding in Altium, measured with an FFT on 1 million samples
- Brought up 3 revisions of a sensor interface board, finding 7 layout faults with an oscilloscope and thermal camera before the design review
- Wrote a Python test harness that checks 64 board signals over SPI in 90 seconds, adopted for production testing of 200 units
- Reduced a power rail's ripple by 60% by adding an LC post-filter sized in LTspice and confirmed on the bench

Embedded Software Intern, Example Robotics, Hillcrest, ST May 2024–Aug 2024

- Ported motor-control firmware from bare-metal C to FreeRTOS on an STM32, keeping the 10 kHz control loop jitter under 2 µs on a logic analyzer
- Added a bootloader that updates firmware over CAN in 40 seconds, tested across 100 update cycles with no failures
- Fixed an encoder overflow bug that caused a 1-in-500 position error, confirmed by a 24-hour run with logged positions

Undergraduate Research Assistant, Example State Power Lab Sep 2023–Apr 2024

- Characterized 12 GaN transistors on a double-pulse tester, measuring switching losses within 5% of datasheet values
- Automated data capture in MATLAB, cutting each test session from 2 hours to 30 minutes

Tutor, Example State Learning Center Jan 2023–May 2023

Projects

Senior Design: Battery Management System for an Electric Kart Fall 2025–Spring 2026
Altium, STM32, C, CAN, LTspice

- Balanced a 24-cell Li-ion pack within 10 mV using passive balancing on a custom 4-layer board, verified over 30 charge cycles
- Kept cell temperature estimates within 1.5 °C of thermocouple readings with a lumped thermal model in C running at 10 Hz
- Shut the pack down within 5 ms of an overcurrent fault, tested with 25 deliberate short-circuit events

Software-Defined Radio FM Receiver Spring 2025
Python, GNU Radio, RTL-SDR, NumPy

- Demodulated broadcast FM in real time at 2.4 MS/s in GNU Radio, with stereo audio confirmed on 6 local stations
- Cut CPU use by 40% by decimating early and replacing two blocks with a NumPy polyphase filter
- Measured 55 dB stereo separation with a test tone, within 5 dB of a commercial receiver on the same antenna

Self-Balancing Robot Fall 2024
Arduino, C++, MPU-6050, MATLAB

- Balanced a two-wheel robot for over 10 minutes using a PID loop tuned from a MATLAB model, recovering from 15° pushes
- Fused gyro and accelerometer data with a complementary filter, keeping tilt error under 1° against a reference inclinometer
- Logged 2,000 samples per run over serial and plotted step responses in MATLAB to check each gain change

Education

B.S. Electrical Engineering, Minor in Computer Science, GPA 3.88/4.00 Spring 2026
Example State University
Relevant coursework: Embedded Systems, Digital Signal Processing, Power Electronics, Control Systems, RF Circuit Design, Computer Architecture, Estimation Theory
Activities: IEEE student branch, treasurer, Tau Beta Pi, Formula SAE electrical team

Skills

Software: C, C++, Python, MATLAB, Verilog, Altium, KiCad, LTspice, FreeRTOS, GNU Radio, Git, Linux
Hardware/Lab: Oscilloscope, Spectrum analyzer, Logic analyzer, Double-pulse tester, Thermal camera, STM32, FPGA
Certifications: Engineer in Training (EIT), IPC J-STD-001 soldering